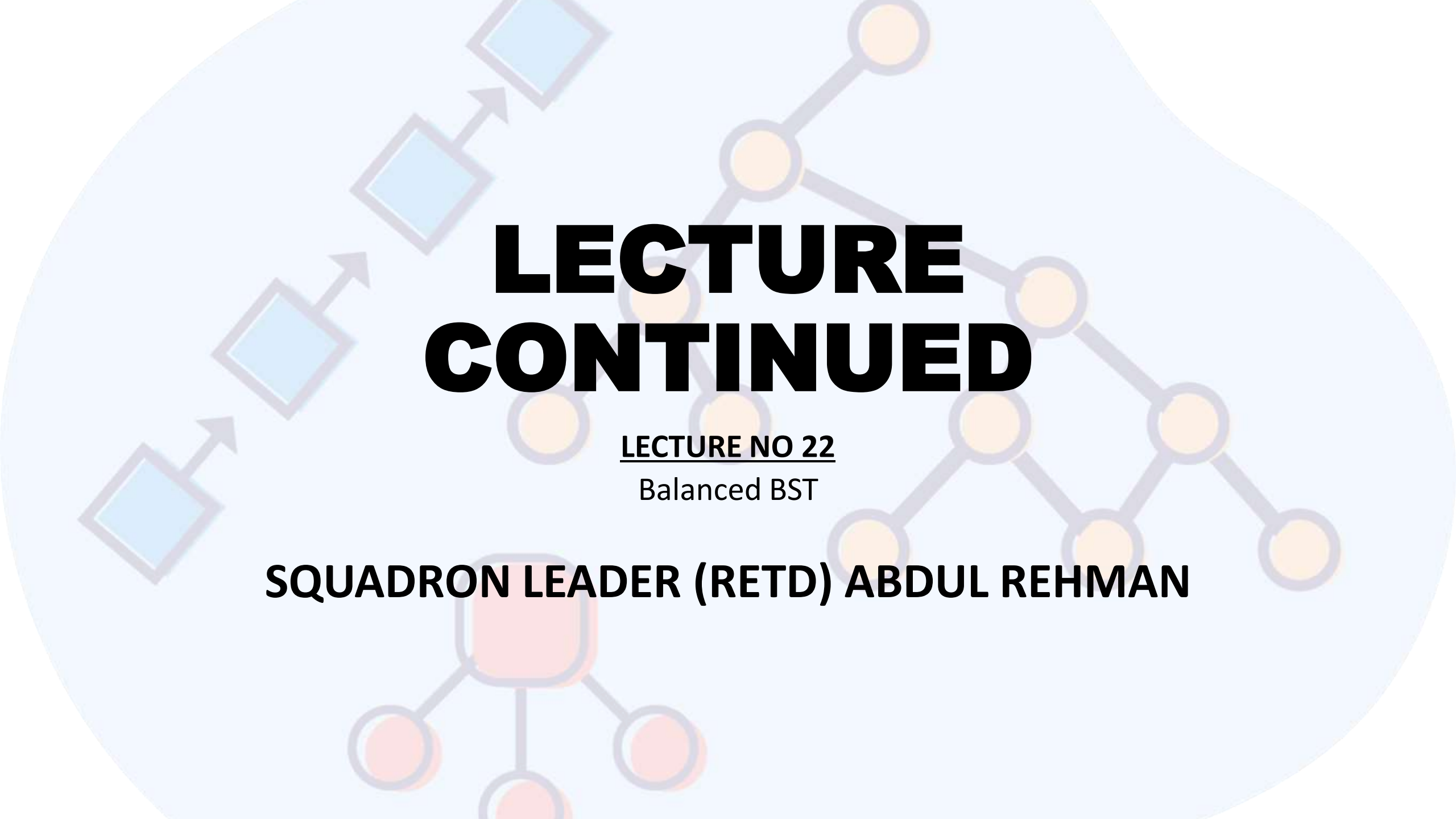




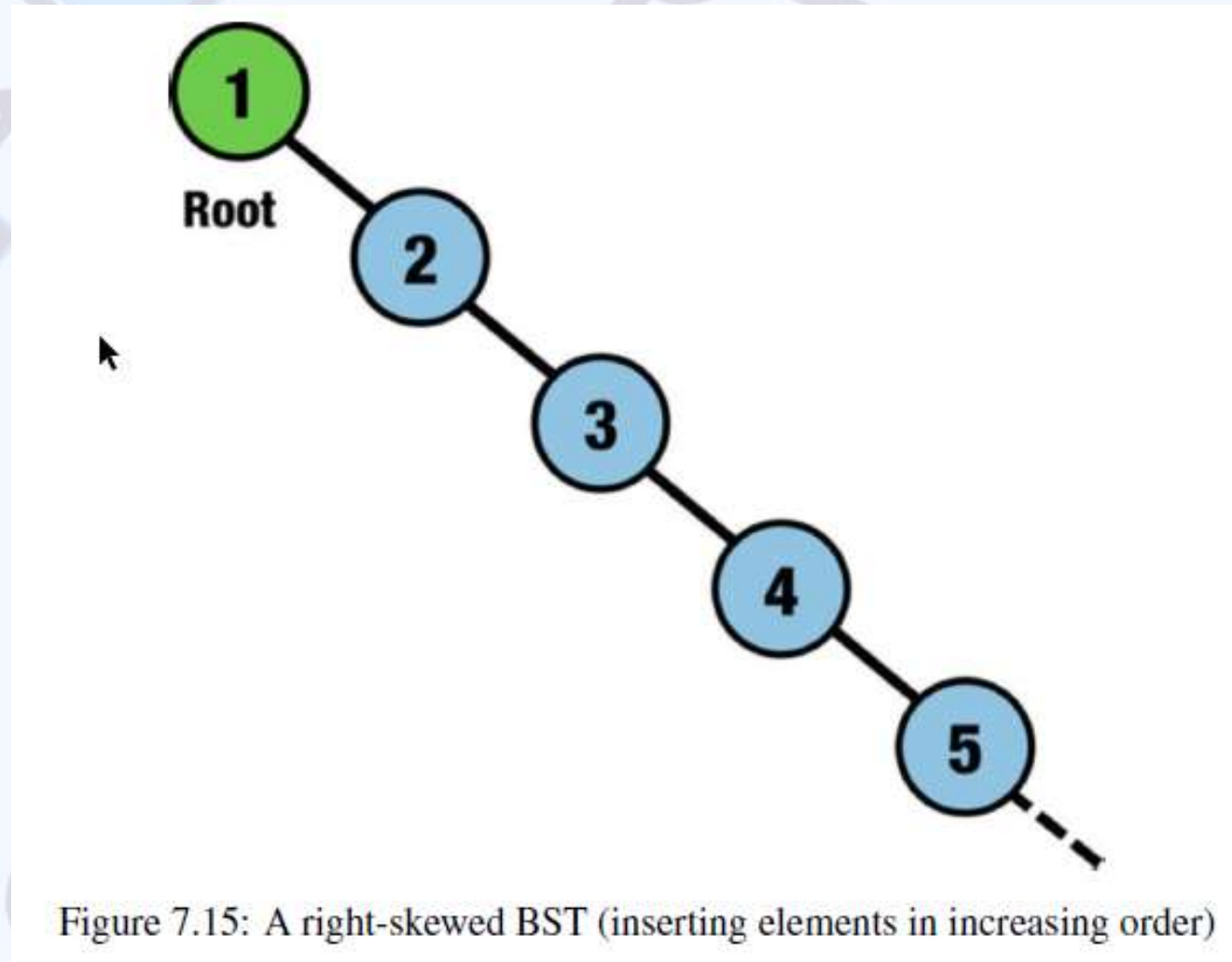
LECTURE CONTINUED

LECTURE NO 22

Balanced BST

SQUADRON LEADER (RETD) ABDUL REHMAN

A RIGHT SKEWED BST



A LEFT SKEWED BST

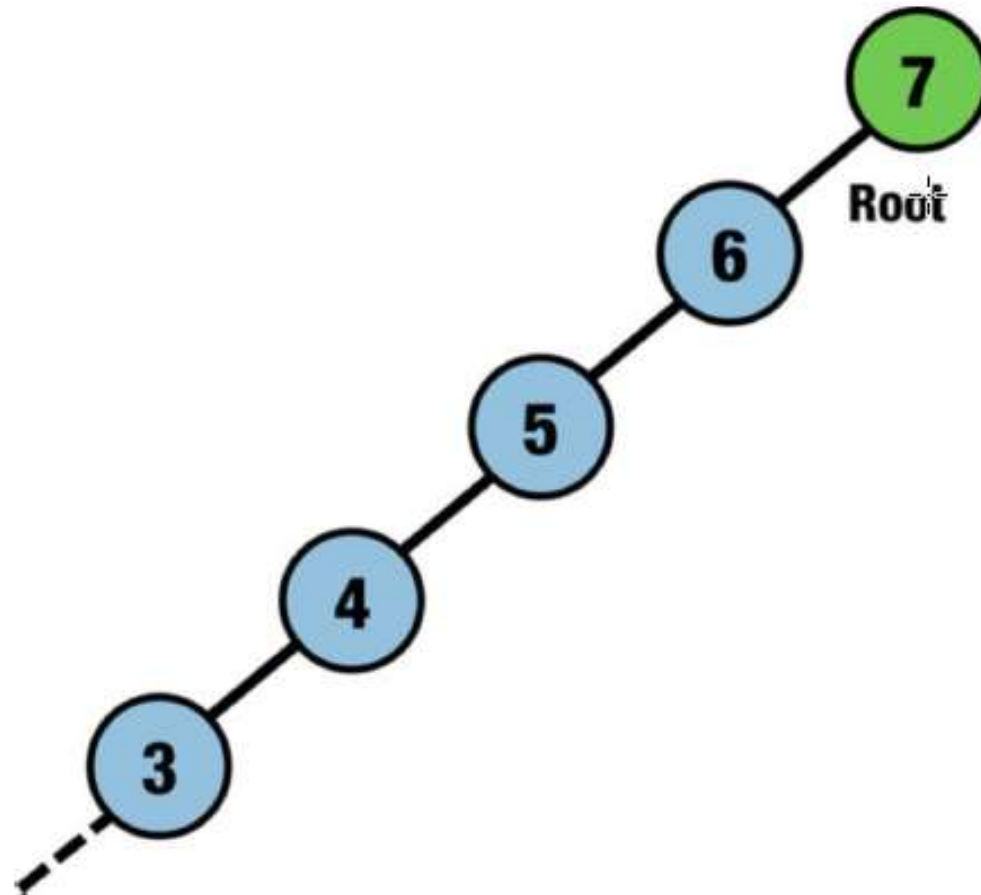


Figure 7.16: A left-skewed BST (inserting elements in decreasing order)

SPARSE BST

- A sparse tree has nodes that are not uniformly distributed leading to significant height differences between subtrees
- This irregular shape can result from random insertions and deletions over time without rebalancing.

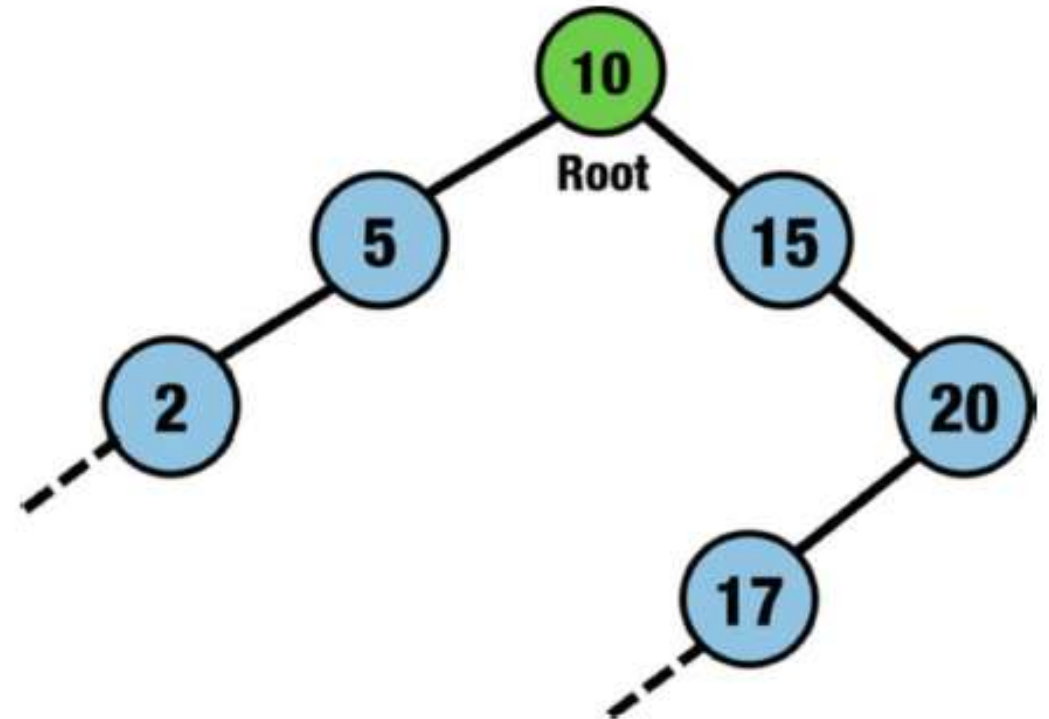


Figure 7.17: A sparse BST with irregular node distribution

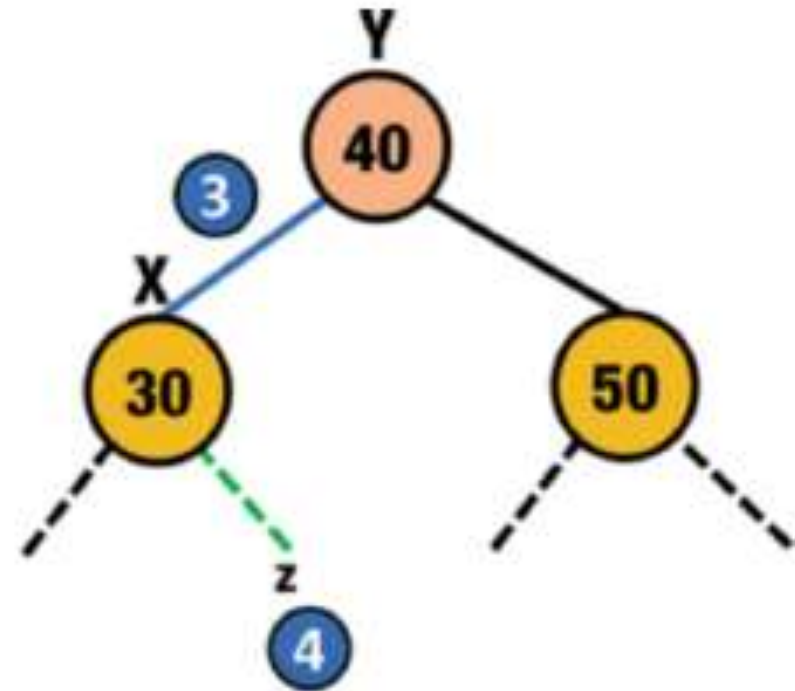
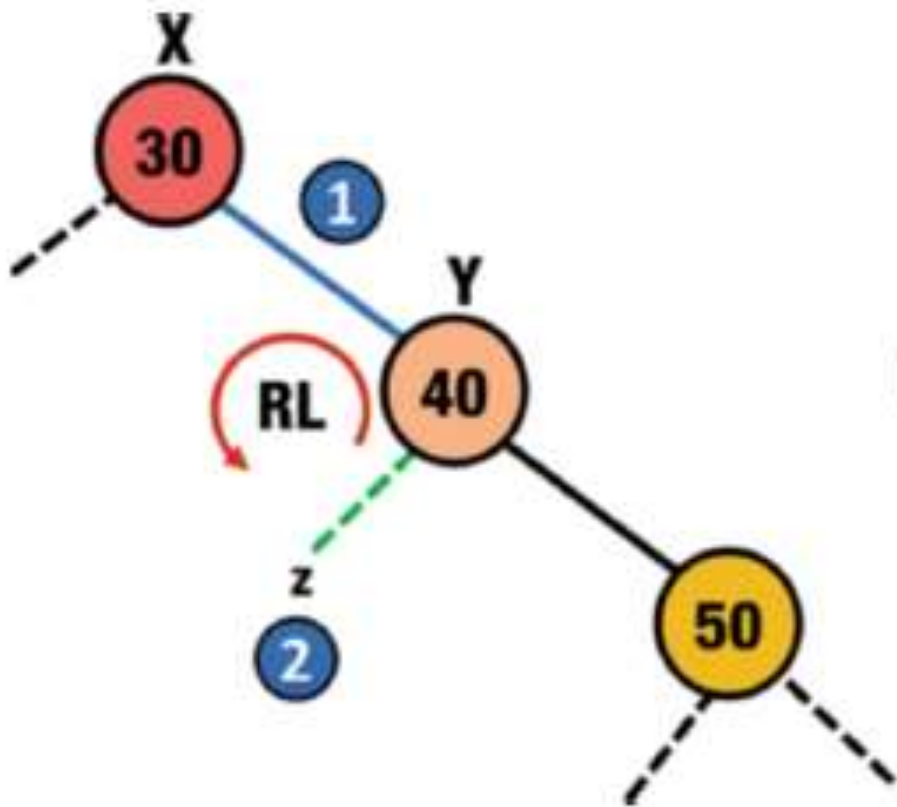
SELF BALANCING TREE - AVL

- The AVL property is defined by the balance factor of a node, which is:
- the difference in height between its right and left subtrees
- The permissible balance factors for a node in an AVL tree are:
 - -1 , 0 , or 1 ,.
- The defining characteristic of AVL trees is
 - For any node N in an AVL tree, **The Absolute Difference** in heights between its left and right subtrees is at most one

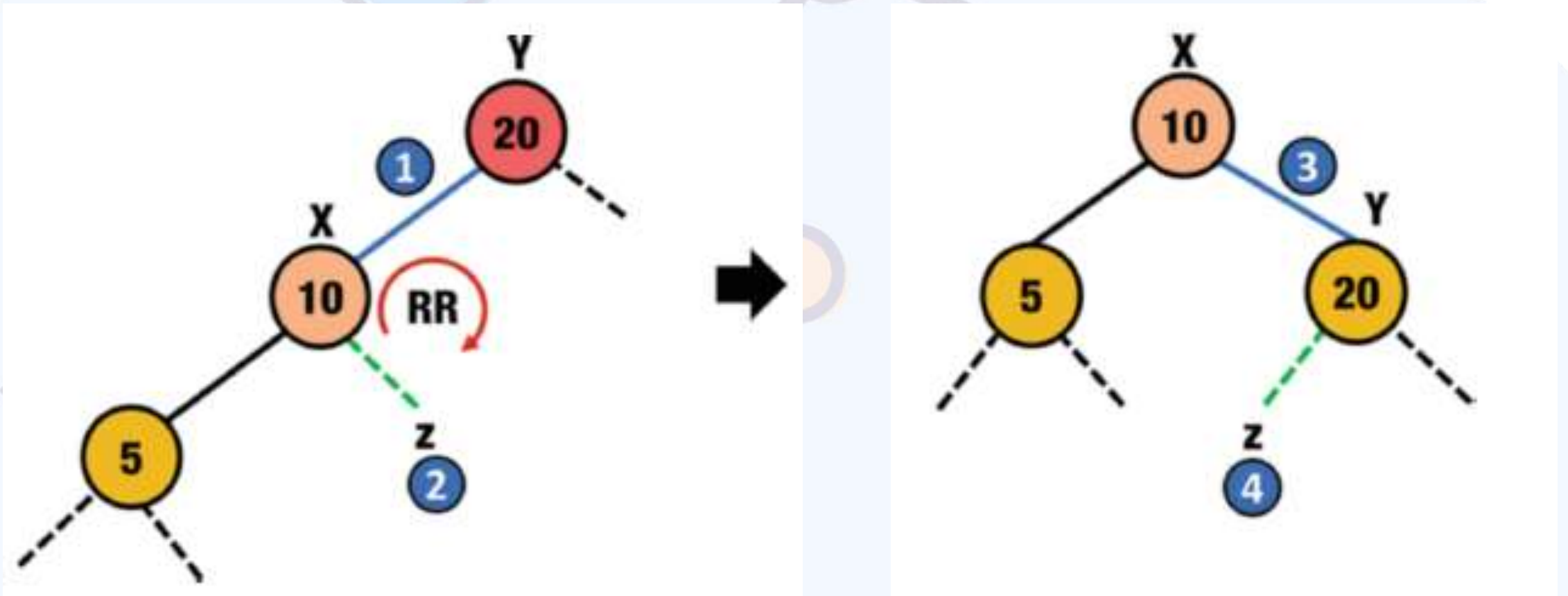
ROTATIONS IN AVL TREES

- There are **four** types of rotations used to rebalance the tree:
 - **Left Rotation:** Applied when the right subtree is taller than the left subtree
 - **Right Rotation:** Applied when the left subtree is taller than the right subtree
 - **Left-Right Rotation:** A combination of left and right rotations used when the left subtree's right child is taller
 - **Right-Left Rotation:** A combination of right and left rotations used when the right subtree's left child is taller

LEFT ROTATION



RIGHT ROTATION



LEFT-RIGHT ROTATION

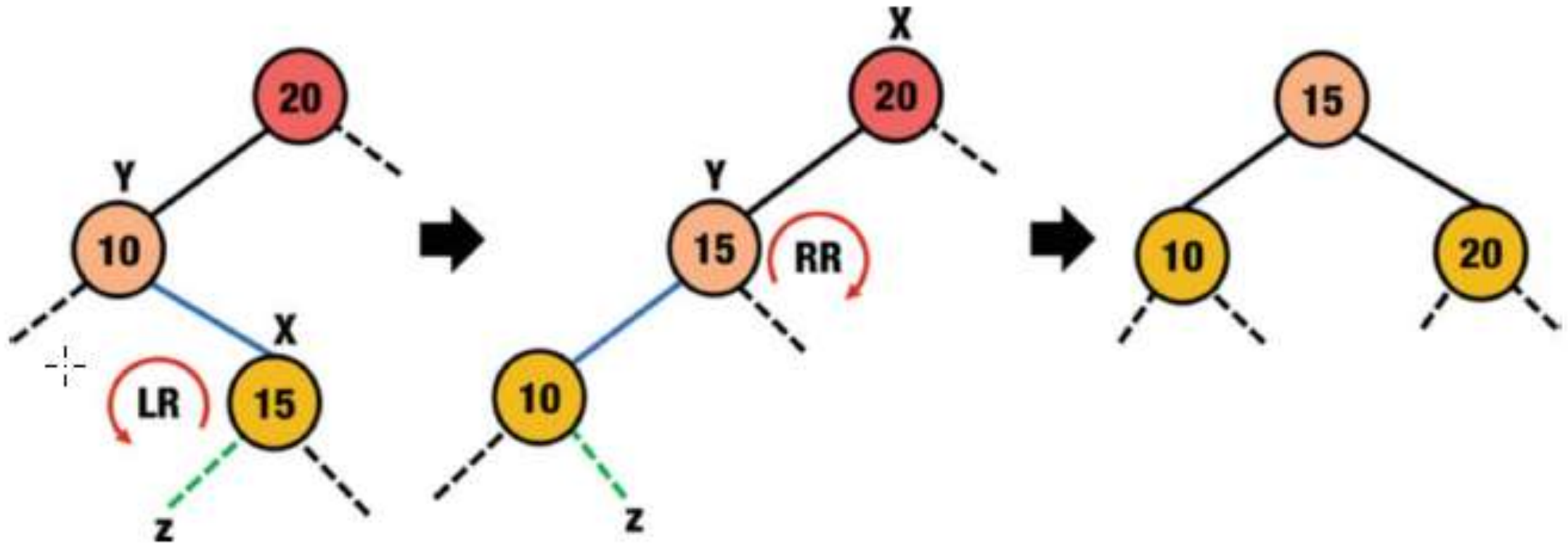


Figure 7.23: Left-right rotation: a combination of left and right rotations used when the left subtree's right child is taller

RIGHT-LEFT ROTATION

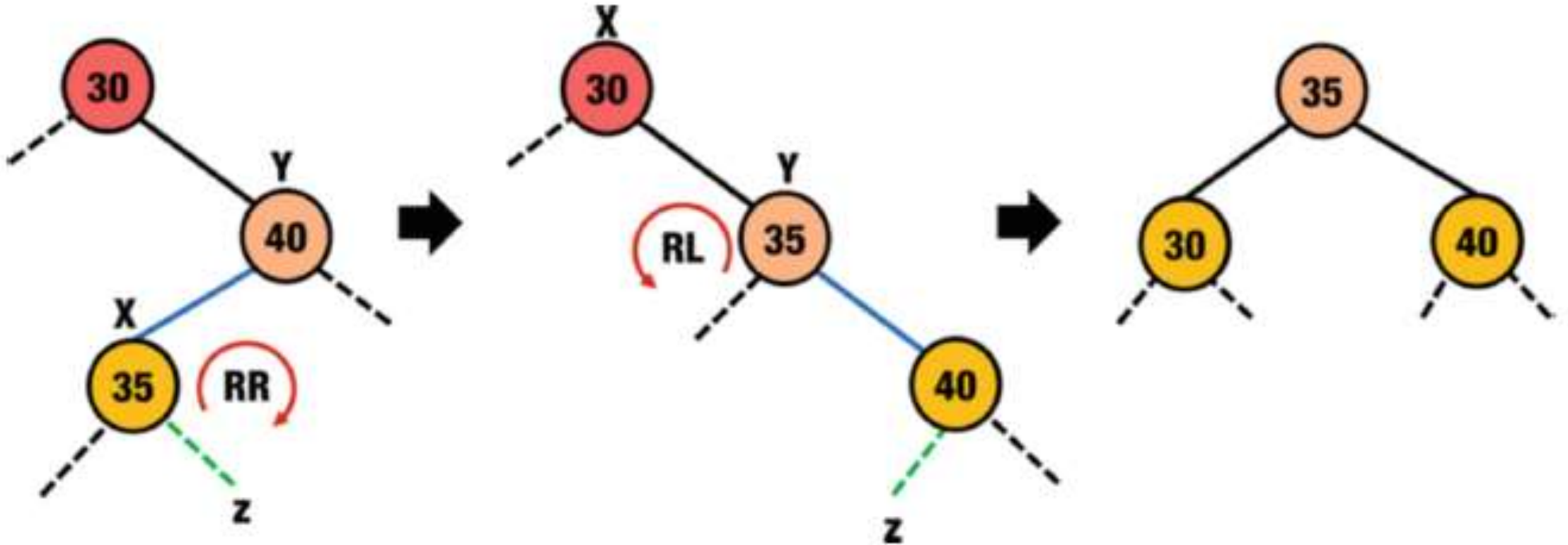
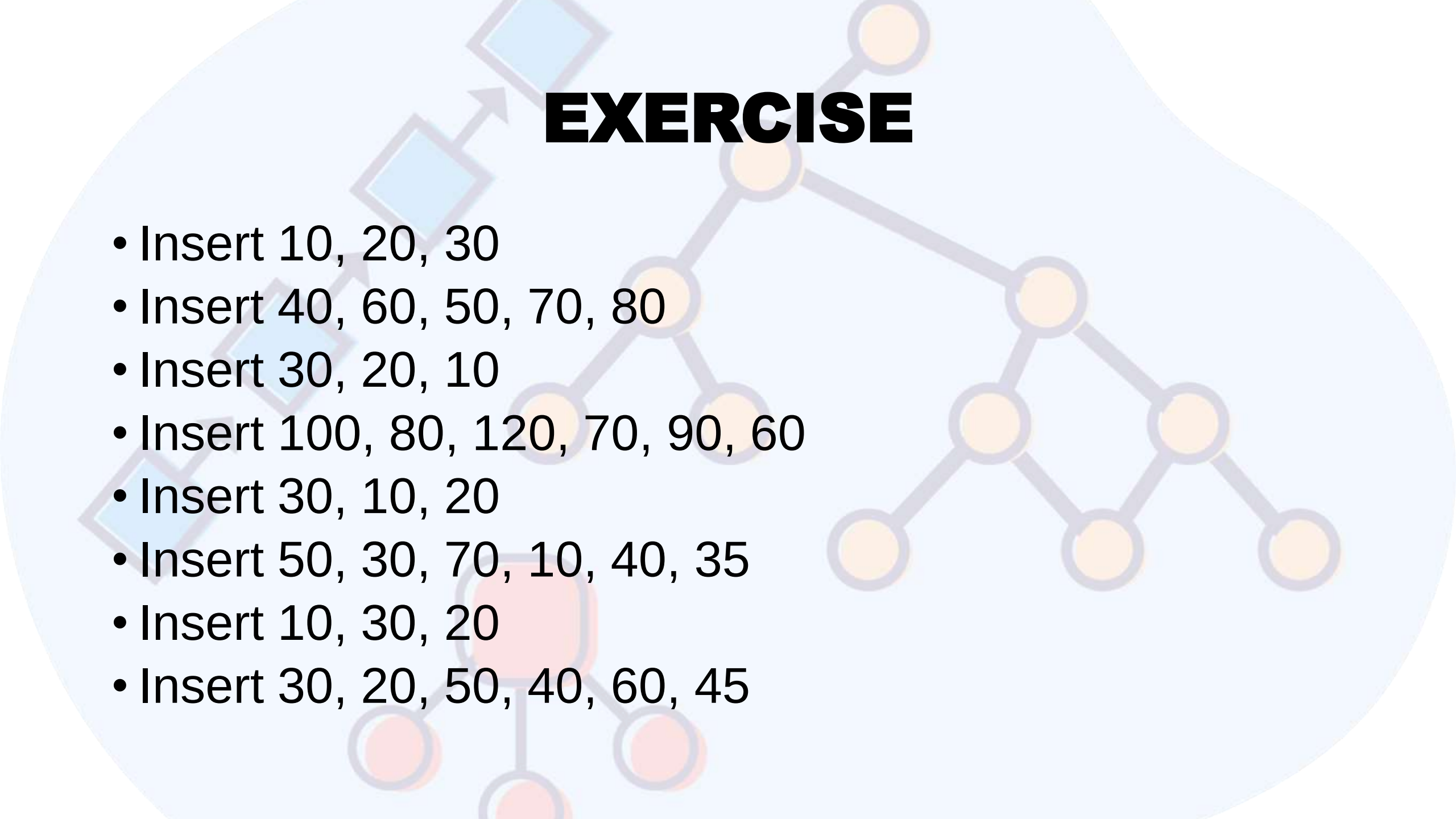


Figure 7.24: Right-left rotation: a combination of right and left rotations used when the right subtree's left child is taller

EXERCISE



- Insert 10, 20, 30
- Insert 40, 60, 50, 70, 80
- Insert 30, 20, 10
- Insert 100, 80, 120, 70, 90, 60
- Insert 30, 10, 20
- Insert 50, 30, 70, 10, 40, 35
- Insert 10, 30, 20
- Insert 30, 20, 50, 40, 60, 45